

PMATH 351 Notes

Weiyu Li

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1 Normed Vector Spaces

Definition 1.1 (Norm). Let V be a vector space. A map $\|\cdot\| : V \rightarrow \mathbb{R}$ is a *norm* if for all $u, v \in V$ and scalars λ ,

1. $\|v\| = 0$ if and only if $v = 0$;
2. $\|\lambda v\| = |\lambda| \|v\|$;
3. $\|u + v\| \leq \|u\| + \|v\|$ (triangle inequality).

A vector space together with a norm is a *normed vector space*.

Example 1.2 (1-norm and supremum norm on $\mathbb{R}^n$). For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$,

$$\|x\|_1 = |x_1| + \dots + |x_n|, \quad \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}.$$

For the 1-norm,

$$\|x + y\|_1 = \sum_{i=1}^n |x_i + y_i| \leq \sum_{i=1}^n |x_i| + \sum_{i=1}^n |y_i| = \|x\|_1 + \|y\|_1.$$

Example 1.3 (Unit balls in $\mathbb{R}^2$). The unit ball for $\|\cdot\|_1$ is the diamond $|x| + |y| \leq 1$, while the unit ball for $\|\cdot\|_\infty$ is the square $\max\{|x|, |y|\} \leq 1$.

Remark 1.4. The same ideas extend to infinite-dimensional spaces. For $p \geq 1$, the standard p -norm is

$$\|(x_1, \dots, x_n)\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

Example 1.5 (Norms on functions). If V is the vector space of real polynomials (and, more generally, suitable function spaces on $[0, 1]$), define

$$\|f\|_1 = \int_0^1 |f(x)| dx, \quad \|f\|_\infty = \sup_{x \in [0,1]} |f(x)|.$$

Theorem 1.6 (Minkowski inequality). Let $p \geq 1$.

1. On the sequence space $\ell^p = \{(x_i) : \sum_i |x_i|^p < \infty\}$,

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{1/p}$$

is a norm.

2. On the space of continuous functions on $[a, b]$,

$$\|f\|_p = \left(\int_a^b |f(x)|^p dx \right)^{1/p}$$

is a norm.

Proof. The nontrivial part is the triangle inequality. Use convexity of $\varphi(x) = x^p$ for $p \geq 1$:

$$\varphi(\alpha x + (1 - \alpha)y) \leq \alpha\varphi(x) + (1 - \alpha)\varphi(y), \quad 0 \leq \alpha \leq 1.$$

For the function-space case, choose

$$\alpha = \frac{\|f\|_p}{\|f\|_p + \|g\|_p}, \quad 1 - \alpha = \frac{\|g\|_p}{\|f\|_p + \|g\|_p}.$$

Then

$$\begin{aligned} \|f + g\|_p^p &= \int_a^b |f + g|^p dx \\ &= (\|f\|_p + \|g\|_p)^p \int_a^b \left| \alpha \frac{f}{\|f\|_p} + (1 - \alpha) \frac{g}{\|g\|_p} \right|^p dx \\ &\leq (\|f\|_p + \|g\|_p)^p \left[\alpha \int_a^b \left| \frac{f}{\|f\|_p} \right|^p dx + (1 - \alpha) \int_a^b \left| \frac{g}{\|g\|_p} \right|^p dx \right] \\ &= (\|f\|_p + \|g\|_p)^p. \end{aligned}$$

Taking p th roots gives Minkowski's inequality. □

Remark 1.7 (Alternate Minkowski proof). An alternate normalized proof uses $\varphi(x) = x^p$ and the same convexity inequality, assuming first $\|f\|_p, \|g\|_p \neq 0$ and treating the zero cases separately.

2 Metric Spaces

Definition 2.1 (Metric). Let X be a nonempty set. A *metric* on X is a function $d : X \times X \rightarrow [0, \infty)$ such that for all $x, y, z \in X$:

1. $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, y) = d(y, x)$;
3. $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is a metric space.

Example 2.2 (Metric from a norm). If $(X, \|\cdot\|)$ is a normed vector space, then

$$d(x, y) = \|x - y\|$$

defines a metric.

Example 2.3 (Weighted graph metric). Let X be the vertex set of a connected graph, with edge weights $w : E \rightarrow (0, \infty)$. Define

$$d(x, y) = \min \left\{ \sum_{j=1}^{n-1} w(x_j, x_{j+1}) : (x_1, \dots, x_n) \text{ is a path from } x \text{ to } y \right\},$$

and $d(x, x) = 0$. This is a metric.

Example 2.4 (Discrete metric). For any nonempty set X ,

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Example 2.5 (p -adic metric on $\mathbb{Q}$). Fix a prime p . Every nonzero $q \in \mathbb{Q}$ can be written $q = p^n a/b$ with $p \nmid a, b$. Define

$$|q|_p = p^{-n}, \quad |0|_p = 0,$$

and

$$d_p(q, r) = |q - r|_p.$$

The key inequality is the ultrametric inequality

$$|q + r|_p \leq \max\{|q|_p, |r|_p\} \leq |q|_p + |r|_p,$$

so d_p is a metric.

Example 2.6 (Product metric on binary sequences). For $X = \{0, 1\}^{\mathbb{N}}$ and $b = (b_j), c = (c_j)$, define

$$d(b, c) = \begin{cases} 0, & b = c, \\ 2^{-n}, & n = \min\{j : b_j \neq c_j\}. \end{cases}$$

Example 2.7 (A series metric). For binary sequences b, c , define

$$\ell(b, c) = \sum_{j=1}^{\infty} \frac{|b_j - c_j|}{2^j}.$$

This metric can be compared with the first-difference metric above, yielding bounds of the form

$$d(b, c) \leq \ell(b, c) \leq 2d(b, c).$$

Definition 2.8 (Relativized metric space). If (X, d) is a metric space and $\emptyset \neq Y \subseteq X$, then $(Y, d|_{Y \times Y})$ is the induced (relativized) metric space.

Definition 2.9 (Equivalent metrics). Metrics d_1, d_2 on X are *equivalent* if there exist $c_1, c_2 > 0$ such that

$$c_1 d_1(x, y) \leq d_2(x, y) \leq c_2 d_1(x, y)$$

for all $x, y \in X$.

Example 2.10 (Equivalence of p -norms on $\mathbb{R}^n$). For $1 \leq p < \infty$,

$$\|x\|_{\infty} \leq \|x\|_p \leq n^{1/p} \|x\|_{\infty}.$$

Hence the metrics induced by $\|\cdot\|_p$ and $\|\cdot\|_\infty$ are equivalent on $\mathbb{R}^n$.

3 Topology of Metric Spaces

Definition 3.1 (Open and closed balls). For $x \in X$ and $r > 0$,

$$B_r(x) = \{y \in X : d(x, y) < r\}, \quad \overline{B}_r(x) = \{y \in X : d(x, y) \leq r\}.$$

Definition 3.2 (Neighborhood, open set, closed set). A set $N \subseteq X$ is a neighborhood of x if there exists $r > 0$ such that $B_r(x) \subseteq N$. A set $U \subseteq X$ is open if it is a neighborhood of each of its points. A set $F \subseteq X$ is closed if $X \setminus F$ is open.

Example 3.3. In $\mathbb{R}$ with the Euclidean metric, open intervals are open. Both $\emptyset$ and $\mathbb{R}$ are open and closed. Singletons are closed.

Example 3.4 (Discrete metric). Every subset of a discrete metric space is open (and hence also closed): for $x \in A$, take any radius $r < 1$, so $B_r(x) = \{x\} \subseteq A$.

Theorem 3.5 (Unions of open sets). An arbitrary union of open sets is open.

Proof. Let $U = \bigcup_{i \in I} U_i$ and $x \in U$. Then $x \in U_j$ for some j . Since U_j is open, $B_r(x) \subseteq U_j \subseteq U$ for some $r > 0$. $\square$

Corollary 3.6. An arbitrary intersection of closed sets is closed.

Proposition 3.7 (Finite intersections). A finite intersection of open sets is open. Infinite intersections need not be open. For example,

$$\bigcap_{n=1}^{\infty} B_{1/n}(0) = \{0\}$$

in $\mathbb{R}^2$, which is not open.

Definition 3.8 (Convergence). A sequence (x_n) in a metric space (X, d) converges to $x \in X$ if

$$\lim_{n \rightarrow \infty} d(x_n, x) = 0.$$

Equivalently, for every $\varepsilon > 0$ there is N such that $n \geq N$ implies $d(x_n, x) < \varepsilon$.

Example 3.9. In $\mathbb{Q}$ with the Euclidean metric, $x_n = 1/n$ converges to 0.

Example 3.10. A rational sequence can have no limit in $\mathbb{Q}$ even if it converges in $\mathbb{R}$; for example, one may take a rational sequence converging to $\pi \notin \mathbb{Q}$.

Example 3.11 (Discrete metric). A sequence in a discrete metric space converges if and only if it is eventually constant.

Example 3.12 (3-adic convergence). In $\mathbb{Q}$ with the 3-adic metric, the sequence $x_n = 3^n$ converges to 0 because $|3^n|_3 = 3^{-n} \rightarrow 0$. Also consider the recursively defined sequence

$$y_n = 2 + 3 + 3^2 + \cdots + 2 \cdot 3^{n-1},$$

which can be rewritten so that $d(-1, y_n) = 3^{-n} \rightarrow 0$; hence $y_n \rightarrow -1$ in the 3-adic metric.

Definition 3.13 (Closure). For $A \subseteq X$, the closure is

$$\text{cl}(A) = \overline{A} = \bigcap_{\substack{F \supseteq A \\ F \text{ closed}}} F,$$

the smallest closed set containing A .

Example 3.14. In $\mathbb{R}$ with the usual metric, $\overline{\mathbb{Q}} = \mathbb{R}$. In $\mathbb{Q}$ with the induced metric, $\mathbb{Q}$ is already closed in itself.

Example 3.15. $\mathbb{Z}$ is not closed in $\mathbb{Q}$ under the 3-adic metric: there is a sequence in $\mathbb{Z}$ having a 3-adic limit in $\mathbb{Q} \setminus \mathbb{Z}$.

Theorem 3.16. Every closed set contains all of its limit points (equivalently, limits of convergent sequences from the set remain in the set).

4 Continuity

Definition 4.1 (Continuity at a point). Let (X, d) and (Y, ρ) be metric spaces. A map $f : X \rightarrow Y$ is continuous at $x_0 \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d(x, x_0) < \delta \implies \rho(f(x), f(x_0)) < \varepsilon.$$

It is continuous if it is continuous at every point.

Example 4.2 (Discrete domain). If X has the discrete metric, every map $f : X \rightarrow Y$ is continuous: take $\delta < 1$, which forces $x = x_0$.

Definition 4.3 (Uniform continuity). $f : X \rightarrow Y$ is uniformly continuous if for every $\varepsilon > 0$ there is $\delta > 0$, independent of the point x , such that

$$d(x, y) < \delta \implies \rho(f(x), f(y)) < \varepsilon.$$

Remark 4.4. Continuity allows δ to depend on the point. Uniform continuity requires a single δ to work everywhere.

Example 4.5 (Continuous but not uniformly continuous). Consider $f : (0, 1) \rightarrow (0, \infty)$, $f(x) = 1/x$. It is continuous. It is not uniformly continuous because points close to 0 can be arbitrarily close while their images stay separated by a fixed amount.

Definition 4.6 (Sequential continuity). $f : X \rightarrow Y$ is sequentially continuous at x if every sequence $x_n \rightarrow x$ satisfies $f(x_n) \rightarrow f(x)$.

Theorem 4.7 (Equivalent characterizations of continuity). For $f : X \rightarrow Y$ between metric spaces, the following are equivalent:

1. f is continuous;
2. for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X ;
3. f is sequentially continuous.

Proof. (1) $\Rightarrow$ (2): Let $x_0 \in f^{-1}(V)$. Since V is open, choose $\varepsilon > 0$ with $B_\varepsilon(f(x_0)) \subseteq V$. Continuity gives $\delta > 0$ such that $B_\delta(x_0) \subseteq f^{-1}(V)$.

(1) $\Rightarrow$ (3): If $x_n \rightarrow x$, then for every $\varepsilon > 0$, continuity gives $\delta > 0$, and eventually $d(x_n, x) < \delta$, so $\rho(f(x_n), f(x)) < \varepsilon$.

(3) $\Rightarrow$ (1): Contrapositively, if f is not continuous at x_0 , then there is $\varepsilon_0 > 0$ such that for every n we can pick x_n with $d(x_n, x_0) < 1/n$ but $\rho(f(x_n), f(x_0)) \geq \varepsilon_0$. Then $x_n \rightarrow x_0$ but $f(x_n) \not\rightarrow f(x_0)$. $\square$

Proposition 4.8 (Operations preserving continuity). 1. Compositions of continuous maps are continuous.
2. Linear combinations of continuous real-valued functions are continuous.
3. Product maps are continuous with standard product metrics.

Definition 4.9 (Isometry). A map $f : X \rightarrow Y$ is an isometry if

$$d(x_1, x_2) = \rho(f(x_1), f(x_2))$$

for all $x_1, x_2 \in X$.

Example 4.10. Rotations, reflections, translations, and compositions of these are Euclidean isometries of $\mathbb{R}^2$.

Definition 4.11 (Lipschitz and bi-Lipschitz). $f : X \rightarrow Y$ is Lipschitz if there exists $C < \infty$ such that

$$\rho(f(x_1), f(x_2)) \leq C d(x_1, x_2).$$

It is bi-Lipschitz if for some $0 < c \leq C < \infty$,

$$c d(x_1, x_2) \leq \rho(f(x_1), f(x_2)) \leq C d(x_1, x_2).$$

Definition 4.12 (Homeomorphism). A homeomorphism is a continuous bijection whose inverse is also continuous.

4.1 Finite-dimensional normed spaces

Definition 4.13 (Equivalent norms). Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent if there exist constants $0 < c < C < \infty$ such that

$$c\|v\|_2 \leq \|v\|_1 \leq C\|v\|_2$$

for all v .

Theorem 4.14. All norms on $\mathbb{R}^n$ are equivalent.

Proof. It suffices to compare an arbitrary norm $\|\cdot\|$ with $\|x\|_1 = \sum_i |x_i|$. Let e_i be the standard basis and

$$C = \max_i \|e_i\|.$$

For $v = \sum_i v_i e_i$,

$$\|v\| \leq \sum_i |v_i| \|e_i\| \leq C \|v\|_1.$$

Now consider the $\|\cdot\|_1$ -unit sphere $S = \{v : \|v\|_1 = 1\}$. Since $\|\cdot\|$ is continuous on the compact set S , it attains a minimum $c > 0$. For $v \neq 0$,

$$\left\| \frac{v}{\|v\|_1} \right\| \geq c,$$

so $\|v\| \geq c\|v\|_1$. Thus the norms are equivalent. $\square$

5 Completeness

Definition 5.1 (Cauchy sequence). A sequence (x_n) in a metric space (X, d) is Cauchy if for every $\varepsilon > 0$ there exists N such that

$$n, m \geq N \implies d(x_n, x_m) < \varepsilon.$$

Example 5.2. In $\mathbb{R}$ with the Euclidean metric, $x_n = 1/n$ is Cauchy. In a discrete metric space a Cauchy sequence is eventually constant.

Example 5.3. In the 3-adic metric, sequences can be Cauchy for reasons very different from Euclidean geometry; powers of 3 and nearby 3-adic expansions provide useful examples.

Theorem 5.4. Every convergent sequence in a metric space is Cauchy.

Proof. If $x_n \rightarrow x^*$, choose N such that $d(x_n, x^*) < \varepsilon/2$ for $n \geq N$. Then

$$d(x_n, x_m) \leq d(x_n, x^*) + d(x^*, x_m) < \varepsilon.$$

□

Example 5.5 (A Cauchy sequence with no limit in the space). In $\mathbb{Q}$ with the Euclidean metric, a rational sequence converging in $\mathbb{R}$ to an irrational number is Cauchy in $\mathbb{Q}$ but has no limit in $\mathbb{Q}$. Likewise, in $(0, 1)$ with the Euclidean metric, a sequence converging to 0 is Cauchy but has no limit in $(0, 1)$.

Definition 5.6 (Complete metric space). A metric space is complete if every Cauchy sequence converges to a point of the space.

Example 5.7. $\mathbb{R}$ with its usual metric is complete. Every discrete metric space is complete.

Definition 5.8 (Banach space). A complete normed vector space is called a Banach space.

Theorem 5.9 (Closed subspaces of complete metric spaces). Let (X, d) be complete and $Y \subseteq X$. Then (Y, d) is complete if and only if Y is closed in X .

Proof. If Y is closed and (y_n) is Cauchy in Y , it is Cauchy in X , hence converges to some $x \in X$. Closedness gives $x \in Y$.

Conversely, if Y is complete and $x_n \in Y$ converges in X to x , then (x_n) is Cauchy. Completeness in Y gives a limit $y \in Y$. Limits in metric spaces are unique, so $x = y \in Y$. Thus Y is closed. □

Theorem 5.10. For $1 \leq p < \infty$, the space ℓ^p is complete.

Coordinatewise proof. Let $x^{(n)} = (x_i^{(n)})_{i \geq 1}$ be a Cauchy sequence in ℓ^p . For each fixed coordinate i ,

$$|x_i^{(n)} - x_i^{(m)}| \leq \|x^{(n)} - x^{(m)}\|_p,$$

so $(x_i^{(n)})_n$ is Cauchy in $\mathbb{R}$. Define

$$x_i = \lim_{n \rightarrow \infty} x_i^{(n)}.$$

For n, m sufficiently large,

$$\sum_{i=1}^J |x_i^{(n)} - x_i^{(m)}|^p < \varepsilon^p.$$

Letting $m \rightarrow \infty$ with J fixed yields

$$\sum_{i=1}^J |x_i^{(n)} - x_i|^p \leq \varepsilon^p.$$

Since this is true for all J ,

$$\sum_{i=1}^{\infty} |x_i^{(n)} - x_i|^p \leq \varepsilon^p.$$

Thus $x^{(n)} \rightarrow x$ in ℓ^p , and $x \in \ell^p$. □

5.1 Completeness of the real numbers

Definition 5.11. A set $S \subseteq \mathbb{R}$ is bounded above if there exists M with $s \leq M$ for all $s \in S$; bounded below is defined analogously. A set is bounded if it is bounded both above and below.

Definition 5.12 (Least upper bound). If S is nonempty and bounded above, M is the least upper bound (supremum) if

1. $s \leq M$ for every $s \in S$;
2. for every $\varepsilon > 0$ there exists $s \in S$ with $M - \varepsilon < s$.

Theorem 5.13 (Least upper bound principle). Every nonempty bounded-above subset of $\mathbb{R}$ has a least upper bound.

Remark 5.14. A decimal/bisection-style construction of the supremum proceeds as follows: first trap the supremum between consecutive integers, then subdivide repeatedly into ten subintervals and keep the first subinterval whose right endpoint remains an upper bound.

Theorem 5.15 (Monotone convergence). Every bounded nondecreasing sequence in $\mathbb{R}$ converges. Its limit is the supremum of the set of its terms.

Theorem 5.16 (Bolzano-Weierstrass). Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Nested-interval proof. Trap the sequence in a bounded interval. Bisect the interval and choose a half containing infinitely many terms. Repeat. This gives nested intervals $[L_k, U_k]$ whose lengths tend to 0, and choose a subsequence term from each interval. The monotone endpoint sequences converge to the same limit, so the selected subsequence converges. □

Lemma 5.17. Every Cauchy sequence in $\mathbb{R}$ is bounded.

Proof. Take $\varepsilon = 1$. There is N such that $|x_n - x_N| < 1$ for $n \geq N$. Thus the tail lies in a bounded interval, and adding the finitely many initial terms keeps the whole sequence bounded. $\square$

Theorem 5.18. $\mathbb{R}$ is complete.

Proof. A Cauchy sequence is bounded, hence has a convergent subsequence $x_{n_k} \rightarrow x^*$. Given $\varepsilon > 0$, choose N so that $|x_n - x_m| < \varepsilon/2$ for $n, m \geq N$ and choose k with $n_k \geq N$ and $|x_{n_k} - x^*| < \varepsilon/2$. Then for $n \geq N$,

$$|x_n - x^*| \leq |x_n - x_{n_k}| + |x_{n_k} - x^*| < \varepsilon.$$

$\square$

5.2 Uniform convergence and complete function spaces

Definition 5.19 (Uniform convergence). For functions $f_n : X \rightarrow Y$, we say $f_n \rightarrow f$ uniformly if for every $\varepsilon > 0$ there exists N such that for all $n \geq N$,

$$\sup_{x \in X} \rho(f_n(x), f(x)) < \varepsilon.$$

Example 5.20. On $[0, 1/2]$, the geometric partial sums

$$f_n(x) = 1 + x + \cdots + x^n = \frac{1 - x^{n+1}}{1 - x}$$

converge uniformly to $f(x) = 1/(1 - x)$ because

$$\sup_{x \in [0, 1/2]} \left| \frac{x^{n+1}}{1 - x} \right| \rightarrow 0.$$

Theorem 5.21 (Uniform limit theorem). A uniform limit of continuous functions is continuous.

Proof. Given $x, y \in X$, choose n so that $\rho(f, f_n) < \varepsilon/3$ uniformly, and then choose δ so that $d(x, y) < \delta$ implies $\rho(f_n(x), f_n(y)) < \varepsilon/3$. Then

$$\rho(f(x), f(y)) \leq \rho(f(x), f_n(x)) + \rho(f_n(x), f_n(y)) + \rho(f_n(y), f(y)) < \varepsilon.$$

$\square$

Theorem 5.22. A uniform limit of bounded functions is bounded.

Theorem 5.23. Let $C_b(X)$ be the continuous bounded real-valued functions on X . If $\mathbb{R}$ is complete, then $(C_b(X), \|\cdot\|_\infty)$ is complete.

Sketch. A Cauchy sequence in $\|\cdot\|_\infty$ is pointwise Cauchy, hence has a pointwise limit f . The Cauchy estimate is uniform, so passing to the pointwise limit shows $f_n \rightarrow f$ uniformly. By the uniform limit theorem, f is continuous; boundedness is also preserved. $\square$

Theorem 5.24 (Weierstrass M-test). If $f_n \in C_b(X)$ and

$$\sum_{n=1}^{\infty} \|f_n\|_{\infty} < \infty,$$

then the series $\sum_n f_n(x)$ converges uniformly to a continuous function.

6 Compactness

Definition 6.1 (Open cover). A family $\{U_i\}_{i \in I}$ is an open cover of $A \subseteq X$ if each U_i is open and

$$A \subseteq \bigcup_{i \in I} U_i.$$

Definition 6.2 (Compactness). A set $A \subseteq X$ is compact if every open cover of A has a finite subcover.

Example 6.3. Every finite subset of a metric space is compact.

Example 6.4. The set $A = \{1/n : n \in \mathbb{N}\} \cup \{0\} \subseteq \mathbb{R}$ is compact. Given an open cover, choose one set containing 0; it contains all $1/n$ for sufficiently large n , and the remaining finitely many points can be covered by finitely many additional sets.

Example 6.5. $A = \{1/n : n \in \mathbb{N}\}$ is not compact; one can construct an open cover with no finite subcover. Likewise, $\mathbb{R}$ is not compact, e.g. the cover $\{(-n, n) : n \in \mathbb{N}\}$ has no finite subcover.

Theorem 6.6. If A is compact in a metric space, then A is closed and bounded.

Proof. For closedness, if $x \notin A$, then for each $a \in A$ choose disjoint small balls about a and x . The balls around points of A form an open cover; choose a finite subcover. Taking the minimum of the corresponding radii gives a ball around x disjoint from A , so A^c is open.

For boundedness, fix $a_0 \in A$. The balls $B_n(a_0)$ cover A . A finite subcover is contained in some largest $B_N(a_0)$, so $A \subseteq B_N(a_0)$. $\square$

Definition 6.7 (Sequential compactness). A is sequentially compact if every sequence in A has a convergent subsequence whose limit lies in A .

Proposition 6.8. A finite set is sequentially compact.

Remark 6.9. If A and B are compact, then $A \cup B$ is compact: take finite subcovers of A and B and combine them.

Definition 6.10 (Induced metric and relative openness). For $A \subseteq X$, the induced metric on A is the restriction of d . A subset $U \subseteq A$ is relatively open if $U = A \cap V$ for some open $V \subseteq X$.

Definition 6.11 (Finite intersection property). A family $\{F_i\}_{i \in I}$ has the finite intersection property (FIP) if every finite subcollection has nonempty intersection.

Definition 6.12 (Total boundedness). A metric space A is totally bounded if for every $\varepsilon > 0$ there is a finite set $F \subseteq A$ such that

$$A \subseteq \bigcup_{x \in F} B_\varepsilon(x).$$

Such an F is an ε -net.

Theorem 6.13 (Borel-Lebesgue characterization for metric spaces). For $A \subseteq X$ with the induced metric, the following are equivalent:

1. A is compact;
2. every family of closed subsets of A with the finite intersection property has nonempty total intersection;
3. A is sequentially compact;
4. A is complete and totally bounded.

Selected implications. (1) $\Rightarrow$ (2): If closed sets F_i have empty total intersection, then their complements form an open cover. A finite subcover gives a finite subfamily with empty intersection, contradicting FIP.

(2) $\Rightarrow$ (3): For a sequence (a_n) in A , let

$$F_n = \overline{\{a_k : k \geq n\}}^A.$$

These closed sets have FIP, hence their intersection is nonempty. Pick a^* in the intersection and choose indices n_k recursively so that $d(a_{n_k}, a^*) < 1/k$.

(3) $\Rightarrow$ (4): Sequential compactness implies completeness: every Cauchy sequence has a convergent subsequence, and then the whole Cauchy sequence converges to the same point. If A were not totally bounded, there would exist $\varepsilon > 0$ and a sequence (x_n) with $d(x_n, x_m) \geq \varepsilon$ for $n \neq m$, so no subsequence could converge.

(4) $\Rightarrow$ (1): Given an open cover, recursively construct a Cauchy sequence by choosing centers of smaller and smaller balls that fail to admit finite subcovers. Completeness yields a limit point f^* . One member of the open cover contains a ball about f^* , and sufficiently small members of the construction lie inside it, giving the desired finite subcover and a contradiction. $\square$

Fact 6.14. Totally bounded and bounded are different notions. In infinite-dimensional settings a set can be bounded but not totally bounded.

Example 6.15. In the product space $X = \{0, 1\}^{\mathbb{N}}$ with the first-difference/product metric, X is compact. Under the sup metric it is bounded but not totally bounded.

Theorem 6.16 (Heine-Borel in $\mathbb{R}^n$). A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

6.1 Countability and separability

Definition 6.17 (Countable). A set is countable if its elements can be listed as a sequence. A set is uncountable if it is not countable.

Example 6.18. $\mathbb{Z}$ is countable. $\mathbb{Q} \cap [0, 1]$ is countable. The set $\{0, 1\}^{\mathbb{N}}$ is uncountable by a diagonal argument: if $x^{(1)}, x^{(2)}, \dots$ were a listing, define a new binary sequence that differs from $x^{(i)}$ in coordinate i .

Proposition 6.19. If $f : A \rightarrow B$ is a bijection, then A is countable if and only if B is countable.

Example 6.20. $\mathbb{R}$ is uncountable. One route maps binary sequences into real numbers by a convergent base expansion.

Definition 6.21 (Dense set and separability). A subset $A \subseteq X$ is dense if $\overline{A} = X$. A metric space is separable if it contains a countable dense subset.

Example 6.22. $\mathbb{R}$ is separable because $\mathbb{Q}$ is countable and dense. A discrete metric space is separable if and only if its underlying set is countable.

Theorem 6.23. Every compact metric space is separable.

Proof. For each n , total boundedness gives a finite $1/n$ -net F_n . The union

$$F^* = \bigcup_{n=1}^{\infty} F_n$$

is countable. Given $x \in X$ and $\varepsilon > 0$, choose n with $1/n < \varepsilon$; some $f \in F_n$ lies within ε of x . Thus F^* is dense. $\square$

6.2 Compactness and continuity

Theorem 6.24. If $f : X \rightarrow Y$ is continuous and X is compact, then $f(X)$ is compact.

Proof. If $\{U_i\}$ is an open cover of $f(X)$, then $\{f^{-1}(U_i)\}$ is an open cover of X . Take a finite subcover and map back to obtain a finite subcover of $f(X)$. $\square$

Theorem 6.25 (Heine-Cantor). If $f : X \rightarrow Y$ is continuous and X is compact, then f is uniformly continuous.

Proof. For each $x \in X$, continuity gives a radius $\delta_x > 0$ such that nearby points have images within $\varepsilon/2$. Compactness yields a finite cover by small balls. Let δ be a suitable positive minimum of the finitely many radii. If $d(x, y) < \delta$, route x and y through the same selected center and use the triangle inequality to obtain $\rho(f(x), f(y)) < \varepsilon$. $\square$

6.3 The middle-thirds Cantor set

Definition 6.26 (Cantor set, version 1). Start with $C_0 = [0, 1]$. Remove the open middle third from every interval to obtain C_1 , then repeat. Let

$$C = \bigcap_{n=0}^{\infty} C_n.$$

Each C_n is compact and the family has the finite intersection property, so C is nonempty and compact.

Definition 6.27 (Cantor set, version 2). The Cantor set consists of points whose ternary expansions use only digits 0 and 2:

$$C = \left\{ \sum_{n=1}^{\infty} \frac{a_n}{3^n} : a_n \in \{0, 2\} \right\}.$$

This description gives an uncountable set.

Definition 6.28 (Cantor set, version 3). The Cantor set is the unique nonempty compact set satisfying

$$C = f_1(C) \cup f_2(C),$$

where the similarity maps are

$$f_1(x) = \frac{x}{3}, \quad f_2(x) = \frac{x+2}{3}.$$

Theorem 6.29 (Universality of the Cantor set). For every compact metric space X , there exists a continuous surjection $f : C \rightarrow X$.

Construction idea. For each scale, choose a finite 2^{-n} -net F_n of X (padding so the number of net points is a power of 2 if convenient). Subdivide the Cantor set into the same number of clopen basic intervals and map each such interval to one net point. This gives locally constant continuous maps $S_n : C \rightarrow X$. The construction can be arranged so that S_n is uniformly Cauchy, hence converges uniformly to a continuous map $S^* : C \rightarrow X$. Nested Cantor intervals force every $x \in X$ to occur as $S^*(c)$ for some $c \in C$, so S^* is surjective. $\square$

7 Compact Subsets of $C(X)$ and Arzela-Ascoli

Assume throughout this section that X is compact and

$$C(X) = \{f : X \rightarrow \mathbb{R} : f \text{ continuous}\}, \quad \|f\|_\infty = \sup_{x \in X} |f(x)|.$$

Because X is compact, every continuous function is bounded, so the uniform norm is finite.

Remark 7.1. If $f_n \rightarrow f$ in $\|\cdot\|_\infty$, then $f_n \rightarrow f$ uniformly. Conversely, uniform convergence is exactly convergence in the uniform norm.

Example 7.2. The set

$$K = \{f_n(x) = x^n : n \in \mathbb{N}\} \subseteq C([0, 1])$$

is not compact in the uniform norm. Pointwise, every subsequence has the same pointwise limit

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1, \end{cases}$$

which is discontinuous; therefore no subsequence can converge uniformly in $C([0, 1])$.

Definition 7.3 (Equicontinuity). Let $K \subseteq C(X)$. We say K is *equicontinuous* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every $f \in K$ and all $a, b \in X$,

$$d(a, b) < \delta \implies |f(a) - f(b)| < \varepsilon.$$

The same δ must work for every function in K .

Proposition 7.4 (Compact subsets are equicontinuous). If X is compact and $K \subseteq C(X)$ is compact in the uniform norm, then K is equicontinuous.

Finite-net argument. Fix $\varepsilon > 0$. Compactness of K gives a finite $\varepsilon/3$ -net $f_1, \dots, f_n$. Since each f_i is uniformly continuous on compact X , there exists $\delta_i > 0$ such that

$$d(a, b) < \delta_i \implies |f_i(a) - f_i(b)| < \varepsilon/3.$$

Let $\delta = \min_i \delta_i$. Given $f \in K$, choose i with $\|f - f_i\|_\infty < \varepsilon/3$. Then for $d(a, b) < \delta$,

$$\begin{aligned} |f(a) - f(b)| &\leq |f(a) - f_i(a)| + |f_i(a) - f_i(b)| + |f_i(b) - f(b)| \\ &< \varepsilon. \end{aligned}$$

□

Theorem 7.5 (Arzela-Ascoli). Let X be a compact metric space and $K \subseteq C(X)$. Then K is compact in the uniform norm if and only if K is closed, bounded, and equicontinuous.

Proof. If K is compact, then K is closed and bounded because $C(X)$ is a metric space, and it is equicontinuous by the preceding proposition.

Conversely, suppose K is closed, bounded, and equicontinuous. Since $C(X)$ is complete, it is enough to show that K is totally bounded. Fix $\varepsilon > 0$. By equicontinuity, choose $\delta > 0$ so that

$$d(x, y) < \delta \implies |f(x) - f(y)| < \varepsilon/3$$

for every $f \in K$. Compactness of X gives a finite δ -net $x_1, \dots, x_n$. Define

$$T : K \rightarrow \mathbb{R}^n, \quad T(f) = (f(x_1), \dots, f(x_n)).$$

The map T is continuous. Boundedness of K implies $T(K)$ is bounded, hence totally bounded in finite-dimensional $\mathbb{R}^n$. Choose finitely many functions $f_1, \dots, f_m \in K$ whose image values approximate $T(K)$ within $\varepsilon/3$ coordinatewise.

For $f \in K$ and any $x \in X$, choose x_i with $d(x, x_i) < \delta$. Then for a suitable f_j ,

$$\begin{aligned} |f(x) - f_j(x)| &\leq |f(x) - f(x_i)| + |f(x_i) - f_j(x_i)| + |f_j(x_i) - f_j(x)| \\ &< \varepsilon. \end{aligned}$$

Thus $\{f_1, \dots, f_m\}$ is a finite ε -net for K . Hence K is totally bounded, and since it is closed in complete $C(X)$, it is complete. Therefore K is compact. $\square$

Example 7.6. Fix $M > 0$. The set of functions $f \in C([0, 1])$ satisfying

$$|f(x) - f(y)| \leq M|x - y|$$

for all x, y is equicontinuous. It is not compact unless it is also uniformly bounded (constant functions already show unboundedness).

Example 7.7. If

$$K = \{f \in C([0, 1]) : |f(x) - f(y)| \leq M|x - y|, |f(0)| \leq L\},$$

then K is equicontinuous and uniformly bounded because

$$|f(x)| \leq |f(x) - f(0)| + |f(0)| \leq M + L.$$

After checking closedness, Arzela-Ascoli gives compactness.

8 Connectedness and Path Connectedness

Definition 8.1 (Disconnected and connected). A subset A of a metric space is disconnected if there exist open sets U, V such that

1. $A \subseteq U \cup V$;
2. $U \cap V = \emptyset$;
3. $A \cap U \neq \emptyset$ and $A \cap V \neq \emptyset$.

A set is connected if it is not disconnected.

Example 8.2. The set $\{0, 1\}$ is disconnected. Any finite set with at least two points is disconnected in the induced Euclidean metric. The integers are disconnected. A discrete metric space with at least two points is disconnected. The middle-thirds Cantor set is disconnected.

Theorem 8.3. The interval $[0, 1]$ is connected.

Supremum proof. Suppose $[0, 1]$ is disconnected by disjoint open sets U, V , with $0 \in U$ and $1 \in V$ without loss of generality. Let

$$C = U \cap [0, 1]$$

and set $c^* = \sup C$. Then $c^* \in [0, 1]$. If $c^* \in U$, openness gives a small interval to the right of c^* still in U , contradicting the definition of c^* . If $c^* \in V$, openness gives a small interval to the left lying in V , contradicting the fact that points of C approach c^* from below. Therefore no such separation exists. $\square$

Theorem 8.4 (Continuous images preserve connectedness). If $f : X \rightarrow Y$ is continuous and X is connected, then $f(X)$ is connected.

Proof. If $f(X)$ were separated by open sets U, V , then $f^{-1}(U)$ and $f^{-1}(V)$ would give a separation of X . $\square$

Corollary 8.5 (Connected subsets of $\mathbb{R}$). A connected subset of $\mathbb{R}$ is an interval, possibly with endpoints included or excluded.

Remark 8.6. Unions and intersections of connected sets are not automatically connected. A useful sufficient condition is: if A and B are connected and $A \cap B \neq \emptyset$, then $A \cup B$ is connected.

Proposition 8.7. If A and B are connected and share a point, then $A \cup B$ is connected.

Definition 8.8 (Connected component). For $x \in X$, the connected component of x is the union of all connected subsets of X that contain x . This extends to connected components of arbitrary subsets.

Example 8.9. For

$$X = \bigcup_{j=0}^{\infty} [2j, 2j+1] \subseteq \mathbb{R},$$

the connected component of a point $x \in [0, 1]$ is $[0, 1]$.

Definition 8.10 (Totally disconnected). A metric space X is totally disconnected if for every distinct $x, y \in X$ there is a separation of X placing x and y in different pieces.

Example 8.11. A discrete metric space is totally disconnected. The middle-thirds Cantor set is totally disconnected. The rationals with the Euclidean metric are totally disconnected: if $a < b$ are rational, choose an irrational $c \in (a, b)$ and separate by $(-\infty, c)$ and (c, ∞) .

Fact 8.12. If a connected component contains more than one point, then the space is not totally disconnected.

Definition 8.13 (Path connected). A metric space X is path connected if for every $x, y \in X$ there exists a continuous map

$$f : [0, 1] \rightarrow X$$

with $f(0) = x$ and $f(1) = y$.

Example 8.14. Every interval is path connected. Every convex subset of a normed vector space is path connected via the line segment

$$f(t) = tx + (1 - t)y.$$

Example 8.15 (Connected but not path connected). A standard oscillatory example is the following: the graph of a function such as $\sin(1/x)$ for $x > 0$, together with a vertical limiting segment at $x = 0$, is connected but not path connected.

9 Cantor Sets and Hausdorff Dimension

Definition 9.1 (Linear Cantor set). Let $n \geq 2$ and let A be a finite digit set, typically $A \subseteq \{0, 1, \dots, n-1\}$. The linear Cantor set is

$$C_{A,n} = \left\{ \sum_{j=1}^{\infty} \frac{a_j}{n^j} : a_j \in A \right\}.$$

Example 9.2. The middle-thirds Cantor set is $C_{\{0,2\},3}$. Every point is an accumulation point of other points of the Cantor set.

Definition 9.3 (Box-counting dimension). For a bounded set $A \subseteq \mathbb{R}^m$, let $N_\varepsilon(A)$ be the minimum number of ε -balls needed to cover A . The upper box dimension is

$$\overline{\dim}_B(A) = \limsup_{\varepsilon \rightarrow 0} \frac{\log N_\varepsilon(A)}{-\log \varepsilon},$$

and the lower box dimension uses $\liminf$. If they agree, the common value is the box dimension $\dim_B(A)$.

Example 9.4 (Cantor set box dimension). At scale 3^{-n} , the middle-thirds Cantor set is covered by 2^n intervals of length 3^{-n} , suggesting

$$\dim_B(C) = \frac{\log 2}{\log 3}.$$

Definition 9.5 (Hausdorff content and Hausdorff measure). Let $A \subseteq \mathbb{R}^m$, $s \geq 0$, and $\delta > 0$. Define

$$\mathcal{H}_\delta^s(A) = \inf \left\{ \sum_i |U_i|^s : A \subseteq \bigcup_i U_i, |U_i| \leq \delta \right\},$$

where $|U_i|$ denotes diameter. The s -dimensional Hausdorff measure is

$$\mathcal{H}^s(A) = \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^s(A).$$

Theorem 9.6 (Jump property). If $s < t$, then

1. $\mathcal{H}^s(A) < \infty$ implies $\mathcal{H}^t(A) = 0$;
2. $\mathcal{H}^t(A) > 0$ implies $\mathcal{H}^s(A) = \infty$.

Proof. If $|U_i| \leq \delta$, then

$$\sum_i |U_i|^t \leq \delta^{t-s} \sum_i |U_i|^s.$$

Taking infima over covers yields

$$\mathcal{H}_\delta^t(A) \leq \delta^{t-s} \mathcal{H}_\delta^s(A).$$

Let $\delta \rightarrow 0$. □

Corollary 9.7. There is at most one critical value d such that $0 < \mathcal{H}^d(A) < \infty$.

Definition 9.8 (Hausdorff dimension). The Hausdorff dimension is

$$\dim_H(A) = \sup\{s : \mathcal{H}^s(A) = \infty\} = \inf\{s : \mathcal{H}^s(A) = 0\}.$$

Example 9.9 (Countable sets have Hausdorff dimension 0). For $A = \mathbb{Q} \cap [0, 1]$, enumerate $A = \{q_1, q_2, \dots\}$. For any $d > 0$ and $\varepsilon > 0$, cover q_i by a ball of diameter chosen so that the sum of d th powers is below ε . Hence $\mathcal{H}^d(A) = 0$ for every $d > 0$, so $\dim_H(A) = 0$.

Fact 9.10. For linear Cantor sets, the box and Hausdorff dimensions agree under the standard self-similar hypotheses.

Fact 9.11. For $A, B \subseteq \mathbb{R}^m$,

$$\dim_H(A \cup B) = \max\{\dim_H(A), \dim_H(B)\}.$$

Fact 9.12. For the Minkowski sum $A + B = \{a + b : a \in A, b \in B\}$,

$$\dim_H(A + B) \leq \dim_H(A) + \dim_H(B).$$

Fact 9.13. If $\emptyset \neq A \subseteq \mathbb{R}^n$, then

$$0 \leq \dim_H(A) \leq n.$$

Example 9.14 (Sums of Cantor sets). Compare $C_{\{0,2\},3} + C_{\{0,2\},3}$ with $[0, 2]$, and consider digit-sum sets such as $C_{\{0,3\},4} + C_{\{0,3\},4}$. Carries in the digit expansions determine whether the sum is an interval, a totally disconnected set, or another Cantor-type set.

9.1 Cantor sets under addition

Theorem 9.15. If

$$\dim_H(C_{A,n}) < \frac{1}{2},$$

then

$$C_{A,n} + C_{A,n} \neq [0, 2].$$

Proof. Using the dimension estimate for Minkowski sums,

$$\dim_H(C_{A,n} + C_{A,n}) \leq 2 \dim_H(C_{A,n}) < 1,$$

whereas $\dim_H([0, 2]) = 1$. □

Exercise 9.16. Let C be a perfect totally disconnected set with $\dim_H(C) < 1/2$. Show that $C + C$ is a perfect totally disconnected set.

Theorem 9.17 (Self-similarity of linear Cantor sets). For $a \in A$, define

$$S_a(x) = \frac{x + a}{n}.$$

Then

$$C_{A,n} = \bigcup_{a \in A} S_a(C_{A,n}).$$

Proof. Write

$$C_{A,n} = \left\{ \frac{a_1}{n} + \frac{a_2}{n^2} + \cdots : a_j \in A \right\}.$$

Grouping by the first digit $a_1 = a$ gives

$$C_{A,n} = \bigcup_{a \in A} \left(\frac{a}{n} + \frac{1}{n} C_{A,n} \right) = \bigcup_{a \in A} S_a(C_{A,n}).$$

□

Theorem 9.18 (Digit-sum criterion). Let $A \subseteq \{0, 1, \dots, n-1\}$ and define

$$A + A = \{a + b : a, b \in A\} = \{0 = b_0 < b_1 < \cdots < b_k = 2n - 2\}.$$

$C_{A,n} + C_{A,n}$ is an interval exactly when the adjacent digit-sums have no gap at scale n ; equivalently,

$$b_i - b_{i-1} \leq 2 \quad \text{for all } i.$$

Larger gaps generate gaps in the Cantor sum.

Definition 9.19 (Cantorval). A *Cantorval* is a compact subset of $\mathbb{R}$ with nonempty interior for which none of its connected components is isolated.

Example 9.20. A Cantorval can be built by modifying the middle-thirds construction and intermittently retaining additional middle intervals so that intervals persist inside a fractal boundary.

Fact 9.21. Depending on the digit set, $C_{A,n} + C_{A,n}$ can be:

1. an interval;
2. a perfect totally disconnected set;
3. a Cantorval.

10 Baire Category Theorem

Definition 10.1 (Nowhere dense). A set $A \subseteq X$ is nowhere dense if

$$\text{int}(\overline{A}) = \emptyset.$$

Example 10.2. In $\mathbb{R}$ with the Euclidean metric, a singleton is nowhere dense, $\mathbb{Z}$ is nowhere dense, $\mathbb{Q}$ is not nowhere dense because $\overline{\mathbb{Q}} = \mathbb{R}$, and the Cantor set is nowhere dense.

Remark 10.3. In a discrete metric space, the only nowhere dense set is $\emptyset$.

Lemma 10.4. If $A \subseteq X$ is nowhere dense, then $(\overline{A})^c$ is open and dense in X .

Proof. It is open because $\overline{A}$ is closed. To see density, let $x \in X$ and $\varepsilon > 0$. Since A is nowhere dense, the ball $B_\varepsilon(x)$ cannot be contained in $\overline{A}$, so it contains a point of $(\overline{A})^c$. $\square$

Definition 10.5 (First category / meagre). A set $A \subseteq X$ is of first category (meagre) if it can be written as a countable union of nowhere dense sets:

$$A = \bigcup_{n=1}^{\infty} K_n.$$

Example 10.6. Every nowhere dense set is first category. The rationals are first category because they are countable, hence a countable union of singletons. In a discrete metric space the only first-category set is $\emptyset$.

Theorem 10.7 (Baire category theorem). A nonempty complete metric space is not of first category in itself.

Corollary 10.8. If a nonempty complete metric space is written

$$X = \bigcup_{n=1}^{\infty} K_n,$$

then at least one closure $\overline{K_n}$ has nonempty interior.

Nested balls proof. Suppose $X = \bigcup_n K_n$ with each K_n nowhere dense. Pick $x_0 \in X$ and a ball $B_{r_0}(x_0)$. Since K_1 is nowhere dense, there is a closed ball $\overline{B}_{r_1}(x_1) \subset B_{r_0}(x_0)$ disjoint from K_1 , with r_1 small. Continue recursively, choosing

$$\overline{B}_{r_n}(x_n) \subset B_{r_{n-1}}(x_{n-1}), \quad \overline{B}_{r_n}(x_n) \cap K_n = \emptyset,$$

and $r_n \rightarrow 0$. Then (x_n) is Cauchy, hence converges to some $x^* \in X$. Nested closed balls force x^* to belong to every $\overline{B}_{r_n}(x_n)$, so $x^* \notin K_n$ for every n , contradicting $X = \bigcup_n K_n$. $\square$

Example 10.9. Baire category implies that $\mathbb{R}$ with the Euclidean metric is not first category in itself.

11 Non-differentiable Functions and Contraction Mapping

11.1 A category argument for non-differentiability

Remark 11.1. A classical continuous nowhere differentiable function can be represented by a rapidly oscillating series. Its precise formula is not needed for the category argument below.

Theorem 11.2. In $C([0, 1])$, the set of functions that are Lipschitz at at least one point is of first category. Consequently, a generic continuous function is nowhere Lipschitz, and hence nowhere differentiable.

Proof outline. For $k \in \mathbb{N}$, let

$$A_k = \{f \in C([0, 1]) : f \text{ is Lipschitz at some point with constant } k\}.$$

Then the set of functions Lipschitz somewhere is $\bigcup_k A_k$. Each A_k is closed: if $f_n \in A_k$ and $f_n \rightarrow f$ uniformly, choose points x_n where f_n has the Lipschitz estimate, pass to a convergent subsequence $x_n \rightarrow x^*$ using compactness of $[0, 1]$, and pass the estimate to the limit.

To show A_k is nowhere dense, approximate any f uniformly by a function with sufficiently sharp small-scale oscillations near every point. Polynomial approximation (Stone-Weierstrass) gives one way to first approximate f and then add a small non-Lipschitz perturbation. Thus each A_k has empty interior. Baire category implies their union cannot equal $C([0, 1])$. $\square$

11.2 Contraction mapping principle

Theorem 11.3 (Contraction mapping principle / Banach fixed-point theorem). Let (X, d) be a complete metric space and suppose $f : X \rightarrow X$ satisfies

$$d(f(x), f(y)) \leq k d(x, y)$$

for some $0 \leq k < 1$. Then:

1. there exists a unique fixed point $x^* \in X$ with $f(x^*) = x^*$;
2. for any $x_0 \in X$, the iteration

$$x_{n+1} = f(x_n)$$

is Cauchy and converges to x^* .

Proof. Let $C = d(x_0, x_1)$. Then

$$d(x_1, x_2) \leq kC, \quad d(x_2, x_3) \leq k^2C, \quad \dots$$

so for $m > n$,

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_{n+1}) + \dots + d(x_{m-1}, x_m) \\ &\leq C(k^n + k^{n+1} + \dots) \\ &= \frac{Ck^n}{1-k} \rightarrow 0. \end{aligned}$$

Thus (x_n) is Cauchy, hence $x_n \rightarrow x^*$ for some $x^* \in X$. Lipschitz maps are continuous, so

$$x^* = \lim x_{n+1} = \lim f(x_n) = f(x^*).$$

If x^*, y^* are fixed points, then

$$d(x^*, y^*) = d(f(x^*), f(y^*)) \leq k d(x^*, y^*),$$

forcing $d(x^*, y^*) = 0$. □

Example 11.4. Consider an iteration such as

$$f(x) = \frac{1}{2}x + \frac{1}{x}$$

on an appropriate domain, where a fixed point solves an algebraic equation. The purpose is to illustrate iterative convergence to a fixed point when a contraction region is available.

Example 11.5 (Logistic map). For

$$f_\lambda(x) = \lambda x(1-x), \quad x \in [0, 1],$$

$$|f_\lambda(x) - f_\lambda(y)| = \lambda|x-y||1-x-y| \leq \lambda|x-y|.$$

Thus for $0 < \lambda < 1$, the map is a contraction on $[0, 1]$ and the unique fixed point is 0. For larger λ , fixed-point behavior changes; this motivates local rather than global contraction arguments.

Theorem 11.6 (Attractive fixed point from a derivative bound). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuously differentiable, let $f(p) = p$, and suppose

$$|f'(p)| < 1.$$

Then there exists an interval $[a, b]$ containing p such that $f([a, b]) \subseteq [a, b]$, f is a contraction on $[a, b]$, and every iteration $x_{n+1} = f(x_n)$ with $x_0 \in [a, b]$ converges to p .

Proof. By continuity of f' , choose $[a, b]$ around p and $k < 1$ such that $|f'(x)| \leq k$ on $[a, b]$. By the mean value theorem,

$$|f(x) - f(y)| \leq k|x - y|.$$

Shrink the interval if necessary so that $f([a, b]) \subseteq [a, b]$, then apply the contraction mapping principle. $\square$

Example 11.7. For the logistic map with $1 < \lambda < 3$, the nonzero fixed point

$$x^* = 1 - \frac{1}{\lambda}$$

has derivative

$$f'(x^*) = 2 - \lambda,$$

so $|f'(x^*)| < 1$. Hence x^* is locally attractive.

11.3 Hausdorff metric and iterated function systems

Definition 11.8 (Hausdorff metric). Let K be the family of nonempty compact subsets of $\mathbb{R}^n$. For $A, B \in K$ define

$$d_H(A, B) = \inf\{\varepsilon > 0 : A \subseteq B_\varepsilon(B), B \subseteq B_\varepsilon(A)\},$$

where

$$B_\varepsilon(A) = \{x : \text{dist}(x, A) < \varepsilon\}.$$

Remark 11.9. Simple interval examples show that d_H measures the smallest radius by which each set must be thickened to contain the other.

Example 11.10 (Cantor-set operator). On compact subsets of $[0, 1]$, define

$$S(A) = \frac{1}{3}A \cup \left(\frac{1}{3}A + \frac{2}{3}\right).$$

The map S is Lipschitz in d_H with constant $1/3$. The unique fixed point is therefore the middle-thirds Cantor set. This gives an iterated-function-system interpretation of the Cantor set.

11.4 Newton's method

Theorem 11.11 (Newton map). Let f be twice continuously differentiable near a root p with

$$f(p) = 0, \quad f'(p) \neq 0.$$

Define the Newton map

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

Then

$$g(p) = p, \quad g'(p) = 0.$$

Proof. Clearly $g(p) = p$. Differentiating,

$$g'(x) = 1 - \frac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2}.$$

At $x = p$, since $f(p) = 0$,

$$g'(p) = 1 - 1 = 0.$$

□

Corollary 11.12 (Local quadratic behavior). Starting sufficiently close to p , Newton iteration is attracted to p . The Taylor expansion of g around p has no linear term:

$$g(x) = p + \frac{g''(p)}{2}(x - p)^2 + \cdots,$$

which explains the quadratic convergence.

12 Completion of Metric Spaces

Definition 12.1 (Completion). A completion of a metric space (X, d) is a complete metric space (Y, ρ) together with a map $J : X \rightarrow Y$ such that

1. J is an isometry:

$$d(x, y) = \rho(J(x), J(y));$$

2. $J(X)$ is dense in Y ;
3. Y is complete.

Theorem 12.2. Every metric space has a completion.

Function-space construction. Fix $x_0 \in X$. Let $C_b(X)$ denote the bounded continuous functions on X with the sup norm; this is complete. For $x \in X$, define

$$f_x(y) = d(x, y) - d(x_0, y),$$

and set

$$J(x) = f_x.$$

The reverse triangle inequality shows f_x is bounded:

$$|f_x(y)| = |d(x, y) - d(x_0, y)| \leq d(x, x_0).$$

Also

$$\|f_x - f_z\|_\infty \leq d(x, z).$$

Taking $y = z$ (or an equivalent extremal choice) gives equality, hence

$$\|J(x) - J(z)\|_\infty = d(x, z).$$

Thus J is an isometry. Let

$$Y = \overline{J(X)} \subseteq C_b(X).$$

As a closed subset of a complete metric space, Y is complete, and $J(X)$ is dense in Y by construction. $\square$

Example 12.3 (Completion of the rationals in the unit interval). The embedding of $\mathbb{Q} \cap [0, 1]$ by distance functions identifies its completion with the interval $[0, 1]$, i.e. with the closure obtained by filling in irrational limit points.

12.1 Completion by Cauchy sequences

Remark 12.4. The standard quotient construction is as follows. Let Z be the set of all Cauchy sequences in X . For two Cauchy sequences (x_n) and (y_n) define the pseudometric

$$\ell((x_n), (y_n)) = \lim_{n \rightarrow \infty} d(x_n, y_n).$$

Declare $(x_n) \sim (y_n)$ when $\ell((x_n), (y_n)) = 0$. Then $Y = Z/\sim$ becomes a metric space, and the map

$$J(x) = [(x, x, x, \dots)]$$

is an isometric dense embedding. The quotient is complete.

Theorem 12.5 (Universal property of completion). Let (Y, ρ) be a completion of (X, d) with embedding $J : X \rightarrow Y$. If $f : X \rightarrow Z$ is uniformly continuous and Z is complete, then there is a unique uniformly continuous map

$$\tilde{f} : Y \rightarrow Z$$

such that

$$\tilde{f}(J(x)) = f(x)$$

for every $x \in X$.

Proof. For $y \in Y$, choose a Cauchy sequence (x_n) in X with $J(x_n) \rightarrow y$. Uniform continuity of f implies $(f(x_n))$ is Cauchy in Z , hence has a limit. Define

$$\tilde{f}(y) = \lim_{n \rightarrow \infty} f(x_n).$$

If a different approximating sequence is chosen, interleave the two sequences or use the fact that their J -images converge to the same point; uniform continuity then shows the two image sequences have the same limit. Thus $\tilde{f}$ is well defined. It extends f , is uniformly continuous, and uniqueness follows from density of $J(X)$. $\square$

Corollary 12.6 (Essential uniqueness). Any two completions of the same metric space are uniquely isometric in a way that identifies the two copies of X . More precisely, if $J_1 : X \rightarrow Y_1$ and $J_2 : X \rightarrow Y_2$ are completions, there is a unique isometry $T : Y_1 \rightarrow Y_2$ with

$$T \circ J_1 = J_2.$$

13 The Real Numbers as an Ordered Field

Definition 13.1 (Field). A field F is a set equipped with addition and multiplication satisfying the usual algebraic axioms: associativity and commutativity of both operations, distributivity, additive and multiplicative identities, additive inverses, and multiplicative inverses for nonzero elements.

Example 13.2. $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{C}$, and $\mathbb{Z}/p\mathbb{Z}$ for prime p are fields. The integers $\mathbb{Z}$ are not a field because nonzero integers need not have multiplicative inverses in $\mathbb{Z}$.

Definition 13.3 (Ordered field). A field F is an ordered field if there is a subset $P \subset F$ of positive elements such that

$$F = P \sqcup \{0\} \sqcup (-P),$$

and whenever $a, b \in P$, both $a + b$ and ab lie in P . Write $a < b$ when $b - a \in P$.

Lemma 13.4. If F is an ordered field, then $1 \in P$.

Proof. Either $1 \in P$ or $-1 \in P$. If $-1 \in P$, closure of P under multiplication gives

$$1 = (-1)(-1) \in P,$$

contradicting the disjoint decomposition $F = P \sqcup \{0\} \sqcup (-P)$. □

Example 13.5 (Ordered field of rational functions). Define an order on the rational-function field by declaring a rational function p/q positive if it is eventually positive for large arguments. Equivalently, the sign is controlled by the ratio of leading coefficients.

Definition 13.6 (Least upper bound property). An ordered field F has the least upper bound property if every nonempty subset $S \subseteq F$ that is bounded above has a least upper bound in F .

Example 13.7. $\mathbb{R}$ has the least upper bound property. $\mathbb{Q}$ does not: for instance

$$S = \{x \in \mathbb{Q} : x^2 < 2\}$$

has no least upper bound in $\mathbb{Q}$.

Definition 13.8 (Archimedean). An ordered field F is Archimedean if for every $x > 0$ there exists $n \in \mathbb{N}$ such that

$$0 < \frac{1}{n} < x.$$

Equivalently, the natural numbers are unbounded above in F .

Example 13.9. $\mathbb{R}$ and $\mathbb{Q}$ are Archimedean. The ordered field of rational functions is non-Archimedean: the variable x dominates every constant natural number in the eventual order.

Theorem 13.10. Let F be an ordered field.

1. $\mathbb{Q}$ embeds in F , and the image of $\mathbb{Q}$ is $P \cup \{0\} \cup (-P)$ restricted to the rational subfield.
2. If F has the least upper bound property, then F is Archimedean.
3. If F is Archimedean and $x < y$, then there exists a rational number m/n such that

$$x < \frac{m}{n} < y.$$

Thus $\mathbb{Q}$ is dense in F .

Proof idea. The rational embedding follows from repeated addition of the multiplicative identity and formation of inverses.

For the second statement, if $\mathbb{N}$ were bounded above, let $x = \sup \mathbb{N}$. Then $x - 1$ is not an upper bound, so for some $n \in \mathbb{N}$, $x - 1 < n$. Hence $x < n + 1$, contradicting that x is an upper bound.

For the density statement, choose n with $1/n < y - x$. Then choose an integer m so that

$$\frac{m-1}{n} \leq x < \frac{m}{n}.$$

The choice of n forces $m/n < y$. □

Remark 13.11 (Dedekind cuts). Dedekind cuts give another construction of the real numbers: a cut is a nonempty proper subset $C \subset \mathbb{Q}$ that is downward closed and has no maximal element. The reals can be constructed as the set of such cuts, with the usual order and arithmetic. This produces a complete ordered field.

14 p -Adic Numbers

Fix a prime p .

Definition 14.1 (p -adic integers). The p -adic integers $\mathbb{Z}_p$ are the completion of $\mathbb{Z}$ with respect to the p -adic norm $|\cdot|_p$.

Definition 14.2 (p -adic numbers). The p -adic numbers $\mathbb{Q}_p$ are the completion of $\mathbb{Q}$ with respect to $|\cdot|_p$.

Theorem 14.3. Let (x_n) be a Cauchy sequence in $\mathbb{Z}$ under the p -adic norm. Then either

1. $x_n \rightarrow 0$ in the p -adic norm, or
2. $|x_n|_p$ is eventually constant.

Proof. If $x_n \not\rightarrow 0$, then there exists a scale p^{-k} that the norms do not eventually cross. Since the sequence is Cauchy, for sufficiently large m, n we have $|x_n - x_m|_p < p^{-k}$. The ultrametric inequality then forces $|x_n|_p = |x_m|_p$ once both norms are larger than the distance between the terms. Thus the norms stabilize. $\square$

Theorem 14.4 (Digit approximation). For every $a \in \mathbb{Z}_p$ and every $k \in \mathbb{N}$, there exists a digit $a_k \in \{0, 1, \dots, p-1\}$ such that the successive truncations approximate a with increasingly small p -adic error. In particular one obtains an expansion

$$a = \sum_{n=0}^{\infty} a_n p^n, \quad a_n \in \{0, 1, \dots, p-1\}.$$

Corollary 14.5. For every $a \in \mathbb{Z}_p$ there are digits $a_0, \dots, a_n \in \{0, 1, \dots, p-1\}$ with

$$|a - (a_0 + a_1 p + \dots + a_n p^n)|_p \leq p^{-(n+1)}.$$

Example 14.6 (3-adic expansion of -1). Use the identity

$$-1 = 2 + 2 \cdot 3 + 2 \cdot 3^2 + 2 \cdot 3^3 + \dots$$

in $\mathbb{Z}_3$. Indeed, the N th partial sum is

$$2 \sum_{j=0}^N 3^j = 3^{N+1} - 1,$$

so its difference from -1 is 3^{N+1} , whose 3-adic norm tends to 0.

Theorem 14.7. $\mathbb{Z}_p$ is compact.

Proof. By definition $\mathbb{Z}_p$ is complete. It remains to show total boundedness. Given $\varepsilon > 0$, choose k with $p^{-k} < \varepsilon$. Every $a \in \mathbb{Z}_p$ can be approximated modulo p^k by one of the finitely many truncations

$$a_0 + a_1 p + \dots + a_{k-1} p^{k-1}, \quad a_j \in \{0, \dots, p-1\}.$$

Thus finitely many p -adic balls of radius p^{-k} cover $\mathbb{Z}_p$. $\square$

Remark 14.8. Addition and multiplication on $\mathbb{Z}_p$ are defined by taking limits of Cauchy representatives and agree with digit arithmetic including carries. The field $\mathbb{Q}_p$, unlike $\mathbb{R}$, is not an ordered field.

15 Polynomial Approximation

Polynomial approximation is motivated by the problem of approximating continuous functions by smoother or differentiable functions.

15.1 Taylor series and interpolation

Remark 15.1 (Taylor approximation). Taylor approximation has the form

$$f(x) \approx \sum_{n=0}^N \frac{f^{(n)}(c)}{n!} (x - c)^n.$$

There are two drawbacks for general approximation theory: it requires many derivatives, and convergence is controlled only inside the radius of convergence.

Definition 15.2 (Lagrange interpolation polynomial). Let $x_0, \dots, x_n$ be distinct points. Define

$$P_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{x - x_j}{x_k - x_j}.$$

Then

$$P_k(x_j) = \begin{cases} 0, & j \neq k, \\ 1, & j = k. \end{cases}$$

For a function f , the interpolation polynomial is

$$P(x) = \sum_{k=0}^n f(x_k) P_k(x),$$

so $P(x_j) = f(x_j)$ for $j = 0, \dots, n$.

Remark 15.3 (Runge phenomenon). Merely increasing the number of equally spaced interpolation nodes need not improve the uniform approximation. A standard example is

$$f(x) = \frac{1}{1 + 25x^2} \quad (-1 \leq x \leq 1),$$

whose high-degree equally spaced interpolants can oscillate strongly near the endpoints.

15.2 Weierstrass approximation theorem

Theorem 15.4 (Weierstrass approximation theorem). Let $f \in C([0, 1])$. For every $\varepsilon > 0$, there exists a polynomial p such that

$$\|f - p\|_{\infty} < \varepsilon.$$

Kernel proof. First reduce to the case $f(0) = f(1) = 0$ by subtracting the linear function joining the endpoint

values. Extend f by 0 outside $[0, 1]$.

For $n \geq 1$, define

$$Q_n(x) = c_n(1 - x^2)^n,$$

where c_n is chosen so that

$$\int_{-1}^1 Q_n(x) dx = 1.$$

Define

$$g_n(x) = \int_{-1}^1 f(x+t)Q_n(t) dt.$$

The proof has three components.

Step 1: estimate c_n . A trigonometric substitution gives

$$\int_{-1}^1 (1 - x^2)^n dx$$

and obtain a lower bound of order $1/n$; equivalently c_n grows at most linearly in n . The exact constant is not important for the convergence argument.

Step 2: concentration of the kernel. For any fixed $\delta > 0$,

$$\int_{\delta \leq |x| \leq 1} Q_n(x) dx \rightarrow 0.$$

Indeed, on $|x| \geq \delta$,

$$(1 - x^2)^n \leq (1 - \delta^2)^n,$$

and the exponential decay dominates the polynomial growth of c_n .

Step 3: g_n is a polynomial. After writing

$$g_n(x) = \int f(u)Q_n(u-x) du,$$

the integrand is a polynomial in x when u is the integration variable, so integration leaves a polynomial in x .

Finally, since f is uniformly continuous and bounded, choose $\delta > 0$ so that

$$|u - x| < \delta \implies |f(u) - f(x)| < \varepsilon/2.$$

Let $|f| \leq M$. Then

$$\begin{aligned} |g_n(x) - f(x)| &= \left| \int_{-1}^1 (f(x+t) - f(x))Q_n(t) dt \right| \\ &\leq \int_{|t| < \delta} |f(x+t) - f(x)|Q_n(t) dt + \int_{|t| \geq \delta} |f(x+t) - f(x)|Q_n(t) dt \\ &\leq \frac{\varepsilon}{2} + 2M \int_{|t| \geq \delta} Q_n(t) dt. \end{aligned}$$

For large n , the second term is below $\varepsilon/2$, uniformly in x . Hence $g_n \rightarrow f$ uniformly. $\square$

Remark 15.5. After the three kernel facts, the final convergence estimate uses compactness of $[0, 1]$ to obtain uniform continuity and boundedness of f .

16 Stone-Weierstrass Theorem

Let X be compact. Then $C(X)$ is a real vector space under pointwise operations.

Definition 16.1 (Algebra). A subset $\mathcal{A} \subseteq C(X)$ is an algebra if

1. $\mathcal{A}$ is a vector subspace;
2. whenever $f, g \in \mathcal{A}$, the pointwise product fg belongs to $\mathcal{A}$.

Example 16.2. The constant functions form an algebra. The polynomials on $[0, 1]$ form an algebra. Even polynomials form an algebra. Differentiable functions form an algebra.

Definition 16.3 (Vector lattice). An algebra $\mathcal{A} \subseteq C(X)$ is a vector lattice if whenever $f, g \in \mathcal{A}$, the pointwise functions

$$\max(f, g), \quad \min(f, g)$$

also belong to $\mathcal{A}$.

Example 16.4. The constant functions form a vector lattice. The set of polynomials is not itself a vector lattice because $\max(f, g)$ need not be polynomial. The uniform closure of the polynomials on $[0, 1]$ is a vector lattice because it is all of $C([0, 1])$ by Weierstrass.

Definition 16.5 (Separates points). An algebra $\mathcal{A} \subseteq C(X)$ separates points if for every distinct $x, y \in X$, there exists $f \in \mathcal{A}$ such that

$$f(x) \neq f(y).$$

Example 16.6. The constant functions do not separate points. Polynomials separate points on an interval. Even functions on a symmetric domain do not separate x and $-x$.

Definition 16.7 (Vanishes at a point). An algebra $\mathcal{A}$ vanishes at $x \in X$ if every $f \in \mathcal{A}$ satisfies $f(x) = 0$.

Example 16.8. The constant functions do not vanish at any point because the constant function 1 is present. The algebra

$$\mathcal{A} = \left\{ \sum_{n=1}^N a_n x^{2n} : a_n \in \mathbb{R} \right\}$$

vanishes at $x = 0$.

Example 16.9 (Polynomial algebra). Let P be the polynomials on $[0, 1]$. Then P is an algebra, is a vector lattice after closure, separates points, and does not vanish at any point because $1 \in P$.

Theorem 16.10 (Stone-Weierstrass, real version). Let X be compact and let $\mathcal{A} \subseteq C(X)$ be an algebra that separates points and does not vanish at any point. Then

$$\overline{\mathcal{A}} = C(X)$$

with closure taken in the uniform norm.

Lemma 16.11 (Closure of an algebra is a vector lattice). If $\mathcal{A} \subseteq C(X)$ is an algebra, then its uniform closure $\overline{\mathcal{A}}$ is a closed algebra and a vector lattice.

Proof. Closedness is immediate. Algebraic closure under linear combinations and products follows by uniform limits and the estimate

$$\|f_n g_n - f g\|_\infty \leq \|f_n(g_n - g)\|_\infty + \|(f_n - f)g\|_\infty.$$

The essential step is to show $|f| \in \overline{\mathcal{A}}$ whenever $f \in \overline{\mathcal{A}}$. By Weierstrass, there are polynomials p_n on a bounded interval containing the range of f such that

$$p_n(t) \rightarrow |t|$$

uniformly. Adjust so $p_n(0) = 0$. If

$$p_n(t) = a_1 t + \cdots + a_k t^k,$$

then

$$p_n(f) = a_1 f + \cdots + a_k f^k \in \overline{\mathcal{A}}$$

because $\overline{\mathcal{A}}$ is an algebra. Uniform convergence gives $p_n(f) \rightarrow |f|$, hence $|f| \in \overline{\mathcal{A}}$. Finally,

$$\max(f, g) = \frac{f + g + |f - g|}{2}, \quad \min(f, g) = \frac{f + g - |f - g|}{2},$$

so both belong to the closure. □

Lemma 16.12 (Two-point interpolation inside the algebra). Let X be compact and let $\mathcal{A} \subseteq C(X)$ be an algebra that separates points and does not vanish anywhere. For any $x \neq y$ in X and any $c, d \in \mathbb{R}$, there exists $f \in \mathcal{A}$ such that

$$f(x) = c, \quad f(y) = d.$$

Construction. Because $\mathcal{A}$ separates x and y , choose $g \in \mathcal{A}$ with

$$g(x) = a, \quad g(y) = b, \quad a \neq b.$$

If $a \neq 0$, use nonvanishing at y to choose $h \in \mathcal{A}$ with $h(y) \neq 0$ and set

$$f(z) = \frac{c}{g(x)} g(z) + \left(\frac{d}{h(y)} - \frac{c h(x)}{g(x) h(y)} \right) h(z),$$

a suitable linear combination after arranging the values appropriately. The degenerate case $a = 0$ is handled by replacing g with a function such as

$$\tilde{g} = g - \frac{g(x)^2}{g(y)}$$

(or an equivalent algebraic modification) to produce a function whose values at x, y allow the same interpolation. The point is that separation plus nonvanishing gives enough two-point freedom. □

Proof of Stone-Weierstrass. Fix $f \in C(X)$ and $\varepsilon > 0$. By the two-point interpolation lemma, for every pair $x, y \in X$ choose $h_{x,y} \in \mathcal{A}$ with

$$h_{x,y}(x) = f(x), \quad h_{x,y}(y) = f(y).$$

For fixed x and each y , define the open set

$$U_{x,y} = \{z \in X : h_{x,y}(z) > f(z) - \varepsilon\}.$$

These sets cover X . By compactness, choose $y_1, \dots, y_m$ such that $U_{x,y_1}, \dots, U_{x,y_m}$ cover X . Let

$$g_x = \max\{h_{x,y_1}, \dots, h_{x,y_m}\} \in \overline{\mathcal{A}}.$$

Then $g_x(z) > f(z) - \varepsilon$ for all z , and $g_x(x) = f(x)$.

Now define

$$V_x = \{z \in X : g_x(z) < f(z) + \varepsilon\}.$$

These open sets cover X . Select a finite subcover $V_{x_1}, \dots, V_{x_N}$ and let

$$g = \min\{g_{x_1}, \dots, g_{x_N}\} \in \overline{\mathcal{A}}.$$

For every $z \in X$,

$$f(z) - \varepsilon < g(z) < f(z) + \varepsilon.$$

Thus $\|f - g\|_\infty < \varepsilon$, proving density. □

17 Best Polynomial Approximation

Let P_n denote the real polynomials of degree at most n on $[0, 1]$ (or on a fixed compact interval).

Definition 17.1 (Best-approximation error). For $f \in C([0, 1])$, define

$$E_n(f) = \inf_{p \in P_n} \|f - p\|_\infty.$$

Theorem 17.2 (Existence of a best approximation). For every $f \in C([0, 1])$ and every n , there exists $p^* \in P_n$ such that

$$\|f - p^*\|_\infty = E_n(f).$$

Proof. Consider a minimizing sequence $p_k \in P_n$. There is no need to consider polynomials with

$$\|p_k - f\|_\infty > \|f\|_\infty,$$

because the zero polynomial already has error $\|f\|_\infty$. Hence restrict to the closed sublevel set

$$S = \{p \in P_n : \|p - f\|_\infty \leq \|f\|_\infty\}.$$

Since P_n is finite dimensional, closed and bounded subsets are compact. The map

$$p \mapsto \|p - f\|_\infty$$

is continuous, so it attains its minimum on S . □

Definition 17.3 (Equioscillation). A continuous function g satisfies the equioscillation property of degree n if there exist

$$x_1 < x_2 < \dots < x_{n+2}$$

such that

$$g(x_i) = (-1)^i \|g\|_\infty$$

for all i , or with all signs simultaneously reversed.

Theorem 17.4 (Equioscillation implies best approximation). Suppose $p \in P_n$ and

$$g = f - p$$

satisfies the equioscillation property of degree n . Then p is a best approximation and

$$\|f - p\|_\infty = E_n(f).$$

Proof. Suppose $q \in P_n$ is strictly better. At each equioscillation point x_i , the difference

$$q(x_i) - p(x_i) = g(x_i) - (f(x_i) - q(x_i))$$

must have the same sign as $g(x_i)$, because the q -error is strictly smaller in magnitude than $\|g\|_\infty$. Thus $q - p$ alternates sign at the $n + 2$ points, so by the intermediate value theorem it has at least $n + 1$ distinct zeros. But $q - p$ has degree at most n , so it must be the zero polynomial. This contradicts strict improvement. $\square$

Theorem 17.5 (Best approximation implies equioscillation). If $p \in P_n$ is a best approximation to f , then $g = f - p$ satisfies the equioscillation property of degree n .

Perturbation argument. At least one point realizes $|g| = \|g\|_\infty$. Suppose there are not $n + 2$ alternating extremal points. Then the extremal set can be grouped into at most $n + 1$ sign blocks. Choose points

$$y_1 < \cdots < y_m, \quad m \leq n,$$

between consecutive blocks and define a polynomial

$$q(x) = C \prod_{j=1}^m (x - y_j),$$

with sign chosen so q has the same sign as g on each extremal block. For sufficiently small $C > 0$, the perturbed polynomial $p + q$ decreases the error at all points where $|g|$ is near the maximum, while changing the error only slightly elsewhere. Hence

$$\|f - (p + q)\|_\infty < \|f - p\|_\infty,$$

contradicting optimality. $\square$

Theorem 17.6 (Uniqueness). The best approximation in P_n is unique.

Proof. Suppose p and q are both best approximations. Then their average is also a best approximation because

$$\left\| f - \frac{p + q}{2} \right\|_\infty \leq \frac{1}{2} \|f - p\|_\infty + \frac{1}{2} \|f - q\|_\infty = E_n(f).$$

Apply equioscillation to $(p + q)/2$. At each of its $n + 2$ extremal points, equality in the triangle inequality forces the errors $f - p$ and $f - q$ to have the same extremal value. Thus $p - q$ has at least $n + 2$ zeros. Since $p - q$ has degree at most n , it must be zero. $\square$

18 Differential Equations and Picard's Theorem

18.1 Basic examples

Remark 18.1. A differential equation involves an unknown function and its derivatives. The order of the equation is the highest derivative that appears.

Example 18.2 (First-order equation). For

$$y'(x) = x^2 + 1,$$

integration gives

$$y(x) = \frac{x^3}{3} + x + C.$$

Example 18.3. The equation

$$(y')^2 + 1 = 0$$

has no real-valued solution because it would require $y' = \pm i$.

Example 18.4 (Second-order equation). The equation

$$y'' = -y$$

has solutions $\sin x$ and $\cos x$, and therefore every linear combination

$$y(x) = a \sin x + b \cos x$$

is also a solution.

18.2 A first contraction argument

Example 18.5. Consider the initial-value problem on $[-1, 1]$

$$y'(x) = -1 + \frac{y(x)}{2}, \quad y(0) = 1.$$

Integrating from 0 to x gives the equivalent integral equation

$$y(x) = 1 - x + \frac{1}{2} \int_0^x y(t) dt.$$

Define $T : C([-1, 1]) \rightarrow C([-1, 1])$ by

$$T(f)(x) = 1 - x + \frac{1}{2} \int_0^x f(t) dt.$$

Then

$$\begin{aligned} \|T(f) - T(g)\|_\infty &= \sup_{x \in [-1, 1]} \left| \frac{1}{2} \int_0^x (f(t) - g(t)) dt \right| \\ &\leq \frac{1}{2} \|f - g\|_\infty. \end{aligned}$$

Thus T is a contraction. By the contraction mapping principle there is a unique fixed point, which is exactly the unique solution to the initial-value problem.

Example 18.6 (Picard iteration). Starting with $f_0 = 0$ and defining $f_{n+1} = T(f_n)$ gives successive approximations. For the preceding example,

$$f_1(x) = 1 - x,$$

and

$$f_2(x) = 1 - x + \frac{1}{2} \int_0^x (1 - t) dt = 1 - \frac{x}{2} - \frac{x^2}{4}.$$

Further iterates produce higher Taylor terms and converge uniformly to the exact fixed point. identify the final closed form and verify directly that applying T returns the same function.

Remark 18.7. The important pattern is not the particular differential equation but the construction of a map $T : C(I) \rightarrow C(I)$ that is both invariant on a suitable set and contractive.

18.3 Global existence under a Lipschitz condition

Definition 18.8 (Lipschitz in the dependent variable). Let I_1, I_2 be intervals. A function

$$\varphi : I_1 \times I_2 \rightarrow \mathbb{R}$$

is Lipschitz in y with constant L if for every fixed $x \in I_1$ and all $y_1, y_2 \in I_2$,

$$|\varphi(x, y_1) - \varphi(x, y_2)| \leq L|y_1 - y_2|.$$

Example 18.9. If $\varphi(x, y) = 1 + x^2$, then it is Lipschitz in y with constant 0. If

$$\varphi(x, y) = -1 + \frac{y}{2},$$

then it is Lipschitz in y with constant $1/2$.

Lemma 18.10 (Iterates become contractive). Suppose φ is Lipschitz in y with constant L . Fix $c_0 \in \mathbb{R}$ and define

$$T(f)(x) = c_0 + \int_c^x \varphi(t, f(t)) dt$$

on $C([a, b])$. If

$$\|f - g\|_\infty \leq M \frac{|x - c|^k}{k!}$$

in the inductive estimate, then

$$|T(f)(x) - T(g)(x)| \leq LM \frac{|x - c|^{k+1}}{(k+1)!}.$$

Consequently,

$$\|T^k(f) - T^k(g)\|_\infty \leq \frac{L^k(b-a)^k}{k!} \|f - g\|_\infty.$$

Proof. Using the Lipschitz condition,

$$\begin{aligned} |T(f)(x) - T(g)(x)| &\leq \int_c^x |\varphi(t, f(t)) - \varphi(t, g(t))| dt \\ &\leq L \int_c^x |f(t) - g(t)| dt. \end{aligned}$$

Insert the inductive bound and integrate to gain one factorial in the denominator. □

Remark 18.11. Since

$$\frac{L^k(b-a)^k}{k!} \rightarrow 0,$$

for large enough k_0 the iterate T^{k_0} is a contraction even when T itself is not.

Lemma 18.12 (Variant with a reference point). Suppose $f, g \in C([a, b])$ satisfy

$$|f(x) - g(x)| \leq M \frac{|x - c|^k}{k!}.$$

Then under the same Lipschitz hypothesis on φ ,

$$|T(f)(x) - T(g)(x)| \leq LM \frac{|x - c|^{k+1}}{(k+1)!}.$$

This is the local form of Picard's theorem.

Theorem 18.13 (Global Picard theorem). Suppose $\varphi(x, y)$ is continuous and globally Lipschitz in y on a rectangle containing $[a, b] \times \mathbb{R}$, with Lipschitz constant L . Then the initial-value problem

$$y'(x) = \varphi(x, y(x)), \quad y(c) = c_0,$$

has a unique solution $y \in C^1([a, b])$.

Proof. Define

$$T : C([a, b]) \rightarrow C([a, b]), \quad T(f)(x) = c_0 + \int_c^x \varphi(t, f(t)) dt.$$

The preceding lemma shows that for sufficiently large k_0 , T^{k_0} is a contraction. Hence T^{k_0} has a unique fixed point f^* . Since

$$T^{k_0}(T(f^*)) = T(T^{k_0}(f^*)) = T(f^*),$$

$T(f^*)$ is also a fixed point of T^{k_0} , so uniqueness implies $T(f^*) = f^*$. Thus f^* is a fixed point of T , and differentiating the integral equation gives the differential equation and initial condition. Uniqueness of a T -fixed point follows from uniqueness for T^{k_0} . $\square$

18.4 Local solutions

Example 18.14. Consider

$$y'(x) = -2xy(x)^2, \quad y(0) = 1.$$

The exact solution is

$$y(x) = \frac{1}{1+x^2}.$$

The function $\varphi(x, y) = -2xy^2$ is not globally Lipschitz in y on all of $\mathbb{R}^2$, so global Picard cannot be applied directly.

Definition 18.15 (Local Lipschitz). A function $\varphi : I \times \mathbb{R} \rightarrow \mathbb{R}$ is locally Lipschitz in y if for every (x, y) there exists a neighborhood rectangle

$$[x - h, x + h] \times [y - k, y + k]$$

on which φ is Lipschitz in y .

Lemma 18.16 (Local Lipschitz becomes uniform on compact convex sets). If φ is locally Lipschitz in y on a compact convex set, then φ is Lipschitz in y on that set with one finite constant L .

Finite-cover chain argument. Cover the compact set by finitely many neighborhoods on which φ has local Lipschitz constants. Let L be the maximum of these constants. For points with the same x -coordinate but separated in y , connect them by a chain of overlapping neighborhoods. On each adjacent pair,

$$|\varphi(x, y_i) - \varphi(x, y_{i+1})| \leq L|y_i - y_{i+1}|.$$

Summing along the chain telescopes to

$$|\varphi(x, y_1) - \varphi(x, y_2)| \leq L|y_1 - y_2|.$$

$\square$

Theorem 18.17 (Local Picard theorem). Suppose φ is continuous and locally Lipschitz in y on a rectangle

$$[a, b] \times [c_0 - R, c_0 + R].$$

Then the initial-value problem

$$y'(x) = \varphi(x, y(x)), \quad y(c) = c_0$$

has a unique solution on some smaller interval $[c - h, c + h]$, where one may choose h small enough that

$$h \leq \frac{R}{\|\varphi\|_\infty}$$

on the compact rectangle (with the obvious interpretation if the sup norm is zero).

Proof. Let

$$T(f)(x) = c_0 + \int_c^x \varphi(t, f(t)) dt$$

on $C([c - h, c + h])$, and restrict T to the closed set

$$C' = \{f \in C([c - h, c + h]) : \|f - c_0\|_\infty \leq R\}.$$

If $h\|\varphi\|_\infty \leq R$, then

$$|T(f)(x) - c_0| \leq \int_c^x |\varphi(t, f(t))| dt \leq h\|\varphi\|_\infty \leq R,$$

so $T(C') \subseteq C'$. The set C' is complete. By compactness and local Lipschitzness, φ has a uniform Lipschitz constant in y on the relevant compact rectangle. As in the global proof, a sufficiently high iterate T^{k_0} is a contraction on C' . Thus T has a unique fixed point in C' , giving the local solution. $\square$

Remark 18.18 (Continuation). A local solution can often be extended by restarting Picard's theorem at an endpoint of the current interval. The extension can stop for a genuine reason, for example because the solution becomes unbounded in finite time.

Remark 18.19 (Systems and higher-order equations). The scalar analysis extends to vector-valued systems. A higher-order scalar differential equation can be rewritten as a first-order system by introducing variables for successive derivatives. For instance, if

$$y'' = F(x, y, y'),$$

set $u_1 = y$ and $u_2 = y'$ to obtain

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}' = \begin{pmatrix} u_2 \\ F(x, u_1, u_2) \end{pmatrix}.$$